Lab 7: Fuzzing for Input Validation Bugs

Screenshot 1: Initial Run on Buggy Application

```
C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7>py photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,8,12
Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|50
Payable Amount: 481.0
```

Screenshot 2: Fuzz Testing with Hypothesis (Bugs Detected & Failures)

```
C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7>py photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,8,12
Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|50
Payable Amount: 481.0

C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7>notepad test_photo_print_order.py

C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7>py -m pytest test_photo_print_order.py
============================= test session starts =============================
platform win32 -- Python 3.14.4, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7
plugins: anyio-4.14.1, hypothesis-6.170.14
collected 3 items

test_photo_print_order.py FFF                                            [100%]

================================== FAILURES ===================================
_____________________________ test_parse_order_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7\test_photo_print_order.py", line 6, in test_parse_order_no_crash
  |     def test_parse_order_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7\test_photo_print_order.py", line 8, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~^^^
    |   File "C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7\photo_print_order.py", line 3, in parse_order
    |     return size.strip().upper(), int(copies)
    |                                  ~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7\test_photo_print_order.py", line 10, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with: {e}"
    | AssertionError: parse_order crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
```

```
=============================== warnings summary ===============================
..\..\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\Rishab Naveen\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================== short test summary info ===========================
FAILED test_photo_print_order.py::test_parse_order_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_photo_print_order.py::test_file_size_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_photo_print_order.py::test_payable_amount_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
========================= 3 failed, 1 warning in 6.54s =========================
```

Screenshot 3: Running Fixed Application

```
C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7>py photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,8,12
Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|50
Payable Amount: 481.0
```

Screenshot 4: Re-running Fuzz Tests on Fixed Code (All Tests Passed)

```
C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7>py -m pytest test_photo_print_order.py
============================= test session starts =============================
platform win32 -- Python 3.14.4, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Rishab Naveen\L6_L7_46_photo_print_order\Lab 7
plugins: anyio-4.14.1, hypothesis-6.170.14
collected 3 items

test_photo_print_order.py ...                                            [100%]

============================== warnings summary ===============================
..\..\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\Rishab Naveen\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed
to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================== 3 passed, 1 warning in 0.97s =========================
```